Bioinformatics algorithms

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Multiple sequence alignment – exercises (**with solutions**)

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1. **Multiple sequence alignment**

   Given the following MSA:

```
-  A  G  G  C  T  A  T  C  A  C  C  T  G
T  A  G  -  C  T  A  C  C  A  -  -  -  G
C  A  G  -  C  T  A  C  C  A  -  -  -  G
C  A  G  -  C  T  A  T  C  A  C  -  G  G
C  A  G  -  C  T  A  T  C  G  C  -  G  G
```

   and the following simple scoring system:

   match score = 1; mismatch penalty = -1; gap penalty = -2

   (a) Compute the total alignment score for the individual columns (1-based position in the alignment) listed in the following table. How many scores do you have to sum for each column?

| | |
|---|---|
| Column 1: | $4 \times \text{gap penalty} + 3 \times \text{mismatch penalty (T with 3 Cs)} + 3 \times \text{match score (Cs)} = -8$ |
| Column 4: | $4 \times \text{gap penalty} + 6 \times \text{gap alignment} (= 0) = -8$ |
| Column 8: | $6 \times \text{mismatch penalty (each C with 3 Ts)} + 4 \times \text{match score } (3 \times \text{Ts},\ 1 \times \text{Cs}) = -2$ |

   **Solution**

   Number of scores to sum in each column for $k = 5$ sequences: $\frac{k(k-1)}{2} = 10$

   (b) Build the alignment profile for this alignment

   (c) Align the following sequence to the existing MSA:[1] `CAGCTTTCACG`

   **Solution**

   In the alignment profile (below, second block) empty fields correspond to a frequency/probability of 0.

[1] This takes long! No need to do the full alignment, just get started! For the exam you would of course not get such a long task, but you should know how to set up the scoring matrix, initialize it and maybe calculate a few entries.

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | - | - | A | G | G | C | T | A | T | C | A | C | C | T | G |
| | - | T | A | G | - | C | T | A | C | C | A | - | - | - | G |
| | - | C | A | G | - | C | T | A | C | C | A | - | - | - | G |
| | - | C | A | G | - | C | T | A | T | C | A | C | - | G | G |
| | - | C | A | G | - | C | T | A | T | C | G | C | - | G | G |
| A | | | 1 | | | | | 1 | | | 0.8 | | | | |
| C | | 0.6 | | | | 1 | | | 0.4 | 1 | | 0.6 | 0.2 | | |
| G | | | | 1 | 0.2 | | | | | | 0.2 | | | 0.4 | 1 |
| T | | 0.2 | | | | | 1 | | 0.6 | | | | | 0.2 | |
| - | 1 | 0.2 | | | 0.8 | | | | | | | 0.4 | 0.8 | 0.4 | |
| - | 0 | -1.6 | -3.6 | -5.6 | -6 | -8 | -10 | -12 | -14 | -16 | -18 | -19.2 | -19.6 | -20.8 | -22.8 |
| C | -2 | (choice: diagonal) $0+\frac{3}{5}(+1)+\frac{1}{5}(-1)$ $+\frac{1}{5}(-2)=0$ left: $-2+\frac{4}{5}(-2)$ top: $-1.6+(-2)$ | | | | | | | | | | | | | |
| A | -4 | (choice: top) $0+(-2)=-2$ left: $-4+\frac{4}{5}(-2)$ diagonal: $-2+\frac{4}{5}(-1)+\frac{1}{5}(-2)$ | | | | | | | | | | | | | |
| G | -6 | | | | | | | | | | | | | | |
| ... | | | | | | | | | | | | | | | |

The final alignment will be:

```
-  A  G  G  C  T  A  T  C  A  C  C  T  G
T  A  G  -  C  T  A  C  C  A  -  -  -  G
C  A  G  -  C  T  A  C  C  A  -  -  -  G
C  A  G  -  C  T  A  T  C  A  C  -  G  G
C  A  G  -  C  T  A  T  C  G  C  -  G  G
C  A  G  -  C  T  T  T  C  A  C  -  -  G
```

2. **Center star algorithm**

Given the following four DNA sequences

(1) ACTGTA
(2) ACTTA
(3) ATCTA
(4) AGTGTA

and the following simple scoring system:

match score = 1; mismatch penalty = -1; gap penalty = -2

(a) For aligning these four sequences using the center star algorithm, which of the four sequences would you pick as the center of the star?

**Solution**

The pairwise global alignments (Needleman-Wunsch) and their alignment scores are:

| | (1) | (2) | (3) | (4) | |
|---|---|---|---|---|---|
| (1) | | ACTGTA<br>*** ** = 3<br>ACT-TA | ACTGTA<br>* *\|** = 1<br>A-TCTA | ACTGTA<br>*\|**** = 4<br>AGTGTA | Total score of (1) = 8 |
| (2) | ACTGTA<br>*** ** = 3<br>ACT-TA | | ACTTA<br>*\|\|** = 1 | ACT-TA<br>*\|* ** = 1<br>AGTGTA | Total score of (2) = 5 |
| (3) | ACTGTA<br>* *\|** = 1<br>A-TCTA | ACTTA<br>*\|\|** = 1<br>ATCTA | | A-TCTA<br>* *\|** = 1<br>AGTGTA | Total score of (3) = 3 |
| (4) | ACTGTA<br>*\|**** = 4<br>AGTGTA | ACT-TA<br>*\|* ** = 1<br>AGTGTA | A-TCTA<br>* *\|** = 1<br>AGTGTA | | Total score of (4) = 6 |

(b) Build the multiple sequence alignment with this center of the star

**Solution**

We just need to add (in arbitrary order) the pairwise alignments of all other sequences with the selected center of the star (highlighted in blue) that we already obtained in the previous step:

ACTGTA
ACT-TA
A-TCTA
AGTGTA

3. **Information content of a sequence motif**

Suppose you have obtained the following **local** multiple sequence alignment (so there are no gaps) of length $w$:

A C T G T A
A C T C T A
A C T C T A
A G T G T A

(a) Compute the alignment profile for the obtained multiple sequence alignment. This essential constitutes the sequence motif of, for example, a transcription factor ...

**Solution**

| | A | C | T | G | T | A |
|---|---|---|---|---|---|---|
| | A | C | T | C | T | A |
| | A | C | T | C | T | A |
| | A | G | T | G | T | A |
| A | 1 | 0 | 0 | 0 | 0 | 1 |
| C | 0 | 0.75 | 0 | 0.5 | 0 | 0 |
| G | 0 | 0.25 | 0 | 0.5 | 0 | 0 |
| T | 0 | 0 | 1 | 0 | 1 | 0 |

(b) Compute the information content $IC(P)$ of this alignment profile/motif $P$

**Solution**

The information content $IC(P)$ is computed as:

$$\sum_{i=1}^{4}\sum_{j=1}^{w} f_{i,j} log_2 \frac{f_{i,j}}{1/4}$$

So we obtain

$$
\begin{aligned}
IC(P) \;=\;& 1 \times log_2 \frac{1}{1/4} + 1 \times log_2 \frac{1}{1/4} && \text{(for A)} \\
&+ 0.75 \times log_2 \frac{0.75}{1/4} + 0.5 \times log_2 \frac{0.5}{1/4} && \text{(for C)} \\
&+ 0.25 \times log_2 \frac{0.25}{1/4} + 0.5 \times log_2 \frac{0.5}{1/4} && \text{(for G)} \\
&+ 1 \times log_2 \frac{1}{1/4} + 1 \times log_2 \frac{1}{1/4} && \text{(for T)} \\
\\
=\;& 1 \times log_2 4 + 1 \times log_2 4 \\
&+ 0.75 \times log_2 3 + 0.5 \times log_2 2 \\
&+ 0.25 \times log_2 1 + 0.5 \times log_2 2 \\
&+ 1 \times log_2 4 + 1 \times log_2 4 \\
\\
=\;& 2 + 2 \\
&+ 1.188722 + 0.5 \\
&+ 0 + 0.5 \\
&+ 2 + 2 \\
\\
=\;& 10.18872
\end{aligned}
$$